Q: 1

Sensor Temperature Analysis

You are given daily temperature readings from **4 weather stations**, each with **5 daily readings**:

```
import numpy as np

data = np.array([[22.5, 19.0, 31.2, 28.7, 25.1],
                 [17.3, 22.8, 30.5, 26.4, 21.9],
                 [33.1, 29.6, 18.4, 24.0, 27.8],
                 [20.2, 23.5, 31.9, 28.1, 22.6]])
```

Task 1 Print the **shape** of `data`. Then compute and print the **mean temperature per station** (one value per row).
Task 2 Using a boolean mask, extract all temperature readings **above 28.0°C** and print them as a 1D array.
Task 3 Normalize the entire `data` array to the range `[0, 1]` using the formula below and print the result rounded to **2 decimal places**:

```
normalized = (data - data.min()) / (data.max() - data.min())
```

Submission Guidelines

Complete all tasks in a **Google Colab Notebook** and submit your link below.

Option A — Google Drive

1. Click **Share** (top-right in Colab)
2. Set access to **Anyone with the link → Viewer**
3. Click **Copy Link** and submit

Option B — GitHub

1. In Colab, go to **File → Download → Download .ipynb**
2. Upload the file to a **new public repository** on github.com
3. Copy the repository URL and submit

⚠ Before submitting, open your link in an **incognito window** to confirm it is publicly accessible and all cell outputs are visible.
Submission Checklist

- Notebook named as `Firstname_Lastname_Assignment.ipynb`
- All tasks completed with visible outputs
- Link is publicly accessible